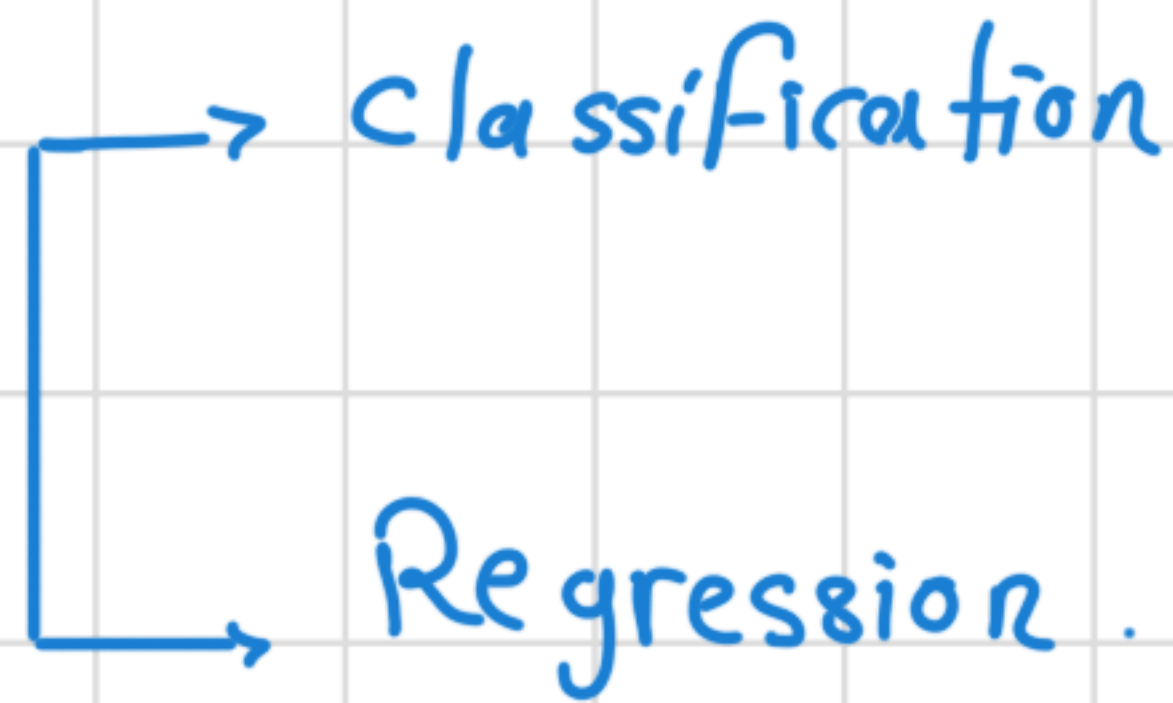


machine learning



supervised ML



1) Regression

predicting
new data

used when the answer you
want is a number

examples:

- Houses prices
- weather

2) Classification

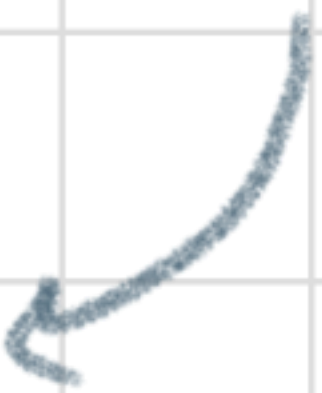
used when the answer is
what category?

Regression

→ To find best fit line
that minimizes the error
between real data points
and the predicted points

Regression $\simeq$ predicting numbers

→ The model learns to put the line "best fit line"
by using a process "gradient Descent"



This adjust the line
until the error between
line and real data points
is as small as possible

Linear Regression Equation

Hypothesis $h_{\theta}(x) = \theta_0 + \theta_1 x$

Parameters θ_0, θ_1

Cost Function $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$

Goal $\longrightarrow$ minimize $J(\theta_0, \theta_1)$

Gradient Descent

$\hookrightarrow$ To get right θ value
That gives the minimum J .

equation of Gradient Descent

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

performance metrics for Regression

R^2



used to check how good the regression model is by comparing the sum of residuals to the total sum.

$R^2 \rightarrow$ has a flaw

(, if unrelated features were added to the model the R^2 will falsely increase.

Adjusted R^2

$\hookrightarrow$ Fixes the flaw of R^2

overfitting

↳ when a model performs extremely well on a training data but fails to perform well on new test data.

underfitting

↳ when the model performs bad on both training data and test data

Ridge Regression

↳ Solves overfitting by adding a penalty parameter to the cost function

which prevents the best fit line from becoming too steep.

Lasso Regression

↳ adds a penalty to prevent overfitting.

mathematical equation

linear equation $y = mx + b$

Dependent variable

slope

Intercept.

→ Multi variate Linear Regression .



used to predict an outcome based
on multiple factors rather
than just one .

